

31. Pattern Matching in Prolog

Aim

To write a Prolog program to demonstrate **pattern matching** using facts and queries.

Algorithm

1. Define facts with different values.
2. Use variables in queries.
3. Prolog matches the query pattern with the facts.
4. Display the matching results.

Program

% Facts

student(101, aryan, cse).

student(102, ravi, ece).

student(103, priya, cse).

student(104, sita, eee).

% Rule for pattern matching

match_department(Name, Department) :-

student(_, Name, Department).

Required Queries

Query 1: Find details of a student

?- student(101, Name, Department).

Output:

Name = aryan,

Department = cse.

Query 2: Find all CSE students

?- student(Id, Name, cse).

Output:

Id = 101, Name = aryan ;

Id = 103, Name = priya.

Query 3: Find the department of Ravi

?- match_department(ravi, X).

Output:

X = ece.

Result

Thus, the **pattern matching concept was successfully implemented in Prolog** by matching queries with facts using variables and constants.

32. Prolog Program to Find the Number of Vowels

Aim

To write a Prolog program to **count the number of vowels in a given list of characters.**

Algorithm

1. Take a list of characters.
2. Check each character.
3. If it is a vowel (a, e, i, o, u), increment the count.
4. Continue until the list is empty.
5. Display the total number of vowels.

Program

% Check whether a character is a vowel

vowel(a).

vowel(e).

vowel(i).

vowel(o).

vowel(u).

% Base case

```
count_vowels([], 0).
```

```
% If head is a vowel
```

```
count_vowels([H|T], Count) :-
```

```
    vowel(H),
```

```
    count_vowels(T, N),
```

```
    Count is N + 1.
```

```
% If head is not a vowel
```

```
count_vowels([H|T], Count) :-
```

```
    \+ vowel(H),
```

```
    count_vowels(T, Count).
```

Sample Input

```
?- count_vowels([h,e,l,l,o], X).
```

Sample Output

```
X = 2.
```

Another Example

```
?- count_vowels([p,r,o,l,o,g], X).
```

```
X = 2.
```

Result

Thus, the **Prolog** program to find the number of vowels was successfully implemented using recursion.